

JOHNSON YEP

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EXPERIENCE

Toyota Boshoku

Jan 2026 - May 2026

Robotics/Automation Engineering Intern

[View Work](#)

- Developed and implemented an end-to-end computer vision and autonomous mobile robot (AMR) system for rack transport. Used **Python** and **OpenCV** to perform real-time rack detection and integrated the system with **Allen-Bradley PLCs** and AMR workflows to automate the movement of **100+** racks per shift.
- Configured and deployed Keyence **computer vision** cameras for defect detection; **preprocessed** and **labeled image datasets** to train AI classification models, improving inspection accuracy and reducing defective parts reaching downstream processes.
- Developed **PLC ladder logic** and **HMI dashboards** for AMRs and production-line machines, delivering real-time status, maps, and analytics that improved troubleshooting, reduced downtime, and supported scalable deployment.

Watonomous

May 2026 - Present

Humanoid Team

[View Work](#)

- Trained a reinforcement learning policy using **PyTorch** in NVIDIA **Isaac Lab** for a SO-ARM101 arm to push objects via contact-only manipulation, using custom reward shaping and a performance-gated curriculum to generalize across randomized spawn positions, and exported the policy to ONNX for **sim-to-real** deployment.
- Integrating a tactile sensor into a humanoid robot hand; simulating a closed-loop control pipeline in **Isaac Lab** that adjusts joint torques over **CAN** bus based on real-time feedback for precision grasping

PROJECTS

Hand Tracking Computer Vision Guitar *Python, OpenCV, MediaPipe*

[View Project](#)

- Created a real-time computer vision pipeline in **Python** that converts hand landmarks from **MediaPipe** into gesture inputs that trigger guitar notes and chords.
- Utilized **OpenCV** for **real-time video processing** and frame analysis to support hand-tracking and gesture detection.

Autonomous Checkers Robot *C++, VEX, Mechanical Design, Embedded Systems*

[View Project](#)

- Built an autonomous checkers-playing robot under strict mechanical and hardware constraints.
- Developed **C++** firmware for board-state representation, sensor-based piece detection, move selection, and overall game control logic.
- Led mechanical design iteration and prototyping of the chassis and claw system, improving stability, alignment, and overall reliability throughout development.

Space Race - 2D Game Development *Java*

[View Project](#)

- Developed a 2D shooter game from scratch in **Java**, designing gameplay, UI, and architecture end-to-end
- Developed core game systems, including player movement, shooting, enemy behavior, collision handling, the main game loop, entity updates, and UI logic for a complete 2D shooter.

CERTIFICATIONS

CS50x: Introduction to Computer Science, Harvard University

[View Certificate](#)

EXTRACURRICULARS

The Boomerang Project - Link Crew

Orillia, Ontario

Student Mentor

Aug 2023 - Jun 2025

- Led orientation team challenges for over **100** incoming grade nine students at Twin Lakes Secondary School, creating a welcoming environment that increased attendance rates by over **30%**.
- Mentored small cohorts of 10 students year-round with a co-leader, increasing first-year participation in clubs/sports and reducing disciplinary referrals by **37%** and suspensions by **20%**.

EDUCATION

University of Waterloo

Waterloo, Ontario

BASc Mechatronics Engineering

April 2030

TECHNICAL SKILLS

- **Languages:** Python, C++, Java
- **ML & Perception:** PyTorch, TensorFlow, Isaac Lab, OpenCV, MediaPipe
- **Embedded & Controls:** CAN Bus, PLC Ladder Logic (Allen-Bradley), HMI Development
- **Developer Tools:** Git, Docker, Linux, Claude Code, Rockwell Automation Software